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Question 1. Make the First Prove

(a)

1. $(p \wedge q) \rightarrow r$ Given
2. $\neg r \vee s$ Given
 - 3.1 $\neg s$ Assumption
 - 3.2 $s \vee \neg r$ Commutativity: 2
 - 3.3 $\neg r$ Elim $\vee$: 3.2, 3.1
 - 3.4 $\neg r \rightarrow \neg(p \wedge q)$ Contrapositive: 1
 - 3.5 $\neg(p \wedge q)$ Modus Ponens: 3.3, 3.4
 - 3.6 $\neg p \vee \neg q$ De Morgan's Law: 3.5
3. $\neg s \rightarrow (\neg p \vee \neg q)$ Direct Proof

(b)

1. $\forall x, (P(x) \wedge Q(x)) \rightarrow R(x)$ Given
2. $\forall x, P(x)$ Given

Let a be arbitrary.

 - 3.1.1 $Q(a)$ Assumption
 - 3.1.2 $P(a)$ Elim $\forall$: 2
 - 3.1.3 $P(a) \wedge Q(a)$ Intro $\wedge$: 3.1.2, 3.1.1
 - 3.1.4 $(P(a) \wedge Q(a)) \rightarrow R(a)$ Elim $\forall$: 1
 - 3.1.5 $R(a)$ Modus Ponens: 3.1.3, 3.1.4
- 3.1 $Q(a) \rightarrow R(a)$ Direct Proof
3. $\forall x, Q(x) \rightarrow R(x)$ Intro $\forall$

Question 2. Pack Up the Proving Van**(a)**

1. $p \vee \neg q$ Given
2. $p \rightarrow r$ Given
3. $q \vee s$ Given
4. $(r \vee s) \rightarrow (v \wedge u)$ Given
 - 5.1 p Assumption
 - 5.2 r Modus Ponens: 5.1, 2
 - 5.3 $r \vee s$ Intro $\vee$: 5.2
5. $p \rightarrow (r \vee s)$ Direct Proof
 - 6.1 $\neg q$ Assumption
 - 6.2 s Elim $\vee$: 3, 6.1
 - 6.3 $r \vee s$: Intro $\vee$: 6.2
6. $\neg q \rightarrow (r \vee s)$ Direct Proof
7. $r \vee s$ Cases: 1, 5, 6
8. $v \wedge u$ Modus Ponens: 7, 4
9. $\neg r \vee (v \wedge u)$ Intro $\vee$: 8

(b)

1. $s \rightarrow (p \vee \neg q)$ Given
2. $(q \wedge p) \vee (s \wedge \neg p)$ Given
3. $q \vee p$ Given
 - 4.1 $s \wedge \neg p$ Assumption
 - 4.2 s Elim $\wedge$: 4.1
 - 4.3 $\neg p$ Elim $\wedge$: 4.1
 - 4.4 $p \vee \neg q$ Modus Ponens: 4.2, 1
 - 4.5 $\neg q$: Elim $\vee$: 4.4, 4.3
 - 4.6 p Elim $\vee$: 3, 4.5
 - 4.7 F Principium Contradictionis: 4.6, 4.3
4. $\neg(s \wedge \neg p)$ Reductio Ad Absurdum
 - 5.1 $q \wedge p$ Assumption
5. $(q \wedge p) \rightarrow (q \wedge p)$ Direct Proof
 - 6.1 $s \wedge \neg p$ Assumption
 - 6.2 F Principium Contradictionis: 6.1, 4
 - 6.3 $q \wedge p$ Ex Falso Quodlibet: 6.2
6. $(s \wedge \neg p) \rightarrow (q \wedge p)$ Direct Proof
7. $q \wedge p$ Cases: 2, 5, 6
8. q Elim $\wedge$: 7

Question 3. Div and Let Div

(a)

Let x , y , and z be arbitrary integers.

- 1.1.1.1.1. $(x \mid y) \wedge (x \mid z)$ Assumption
- 1.1.1.1.2. $(x \mid y)$ Elim $\wedge$: 1.1.1.1.1
- 1.1.1.1.3. $(x \mid z)$ Elim $\wedge$: 1.1.1.1.1
- 1.1.1.1.4. $\exists q_y (y = q_y x)$ Def Divides: 1.1.1.1.2
- 1.1.1.1.5. $\exists q_z (z = q_z x)$ Def Divides: 1.1.1.1.3
- 1.1.1.1.6. $y = n_y x$ Elim $\exists$: 1.1.1.1.4
- 1.1.1.1.7. $z = n_z x$ Elim $\exists$: 1.1.1.1.5
- 1.1.1.1.8. $y + z = n_y x + n_z x$ Algebra
- 1.1.1.1.9. $y + z = (n_y + n_z)x$ Algebra
- 1.1.1.1.10. $\exists n (y + z = nx)$ Intro $\exists$: 1.1.1.1.9
- 1.1.1.1.11. $x \mid (y + z)$ Undef Divides: 1.1.1.1.10

1.1.1.1. $((x \mid y) \wedge (x \mid z) \rightarrow (x \mid (y + z)))$ Direct Proof

1.1.1. $\forall c ((x \mid y) \wedge (x \mid c) \rightarrow (x \mid (y + c)))$ Intro $\forall$

1.1. $\forall b \forall c ((x \mid b) \wedge (x \mid c) \rightarrow (x \mid (b + c)))$ Intro $\forall$

1. $\forall a \forall b \forall c ((a \mid b) \wedge (a \mid c) \rightarrow (a \mid (b + c)))$ Intro $\forall$

(b)

Let x , y , and z be arbitrary integers. Suppose $x \mid y$ and $x \mid z$. By the definition of divides, know $y = n_y x$ and $z = n_z x$ for some integers n_y and n_z . Next, $y + z = n_y x + n_z x = (n_y + n_z)x$. Since addition is closed under the integers, and there is an integer $n = n_y + n_z$ such that $y + z = nx$, we know that $x \mid (y + z)$ by the definition of divides. Therefore, since x , y , and z were arbitrary, the claim holds.

Question 4. Optional Practice Problems**(a)**

1. $p \vee q$ Given
2. $r \wedge \neg p$ Given
3. $\neg(q \wedge \neg s)$ Given
4. r Elim $\wedge$: 2
5. $\neg p$ Elim $\wedge$: 2
6. q Elim $\vee$: 1, 5
7. $\neg\neg q$ Double Negation: 6
8. $\neg q \vee \neg\neg s$ De Morgan's Law: 3
9. $\neg\neg s$: Elim $\vee$: 8
10. s : Double Negation 9
10. $s \wedge r$ Intro $\wedge$: 10, 4

(b)

1. $\exists x P(x)$ Given
2. $\forall x R(x, c)$ Given
3. $\forall x (P(x) \rightarrow R(c, x))$ Given
4. $P(n)$ Elim $\exists$: 1
5. $P(n) \rightarrow R(c, n)$ Elim $\forall$: 3
6. $R(c, n)$ Modus Ponens: 4, 5
7. $R(n, c)$ Elim $\forall$: 2
8. $R(c, n) \wedge R(n, c)$ Intro $\wedge$: 6, 7
9. $\exists y (R(c, y) \wedge R(y, c))$ Intro $\exists$: 8
10. $\exists x \exists y (R(x, y) \wedge R(y, x))$ Intro $\exists$: 9

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(c)**i.** Let x and y be arbitrary odd integers.By the definition of odd, $x = 2n + 1$ and $y = 2m + 1$ for some integers n and m .Then, $x \cdot y = (2n + 1)(2m + 1) = 4nm + 2n + 2m + 1 = 2(2nm + n + m) + 1$.Since addition is closed under the integers, and there is an integer $z = 2nm + n + m$ such that $x \cdot y = 2z + 1$, $x \cdot y$ is odd by the definition of odd.Therefore, since x and y were arbitrary, the claim holds.**ii.** Let x and y be arbitrary rational numbers.By the definition of rational, $x = \frac{a}{b}$ and $y = \frac{c}{d}$ for some integers a, b, c , and d .Then, $x + y = \frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$.Since addition and multiplication are closed under the integers, and there are two integers $p = ad + bc$ and $q = bd$ such that $x + y = \frac{p}{q}$, $x + y$ is rational by the definition of rational.Therefore, since x and y were arbitrary, the claim holds.**iii.** Let a be an arbitrary positive integer.We will show that a satisfies the definition of $\gcd(a, a)$.First, we can see that $a = 1a$. Thus, $a \mid a$ by the definition of divides. It remains to be shown that every positive integer that divides a is no bigger than a .Let b be an arbitrary non-negative integer. Suppose $b \mid a$. Then by definition of divides, $a = qb$. We continue by cases:Suppose that $q = 0$. Then $a = 0b = 0$, which contradicts the first fact that a is a positive integer. Since false is true, anything is true. In particular, we have $b \leq a$.Suppose that $q > 0$. Then $a = qb$, but a cannot be smaller than b by property of integers. Again, $b \leq a$.Since these cases are exhaustive, we have shown that $b \leq a$ holds in general.Since b was arbitrary, we have proven $\gcd(a, a) = a$ by the definition of GCD.Therefore, since a was arbitrary, the claim holds.